Lecture 4: Convex conjugates, and Marginal duality

Lecture 3 showed strong duality first in the polyhedral world of linear programming. Lecture 4 now passes from linear inequality certificates to the duality of convex functions and their epigraphs. The main geometric idea is that the conjugate records affine functions lying below f. For each $\xi \in E^*$, the affine function

$$x \mapsto \langle \xi, x \rangle - f^*(\xi)$$

is the tightest such affine lower bound with slope ξ. Therefore the biconjugate

$$f^{**}(x) = \sup_{\xi \in E^*} \{\langle \xi, x \rangle - f^*(\xi)\}$$

is the supremum of all affine functions lying below f. Geometrically, those affine lower bounds are exactly the affine halfspaces that contain $\mathrm{epi}(f)$, so Fenchel–Moreau will be proved below from the fact that a closed convex set is the intersection of its supporting halfspaces.

4.1 Conjugates and Biconjugation

Definition 4.1 (Convex conjugate and biconjugate)**.** Let $f : E \to \mathbb{R} \cup \{+\infty\}$ be proper. Its convex conjugate is defined by

$$\forall \xi \in E^*, \qquad f^*(\xi) := \sup_{x \in E} \{\langle \xi, x \rangle - f(x)\}.$$

Under the present convention, this keeps f^* valued in $\mathbb{R} \cup \{+\infty\}$. If one allowed an improper function such as $f \equiv +\infty$, then the same formula would give $f^* \equiv -\infty$, which lies outside the current extended-value convention. Let

$$\iota_E : E \to E^{**}, \qquad (\iota_E(x))(\xi) := \langle \xi, x \rangle.$$

Since E is finite-dimensional, ι_E is an isomorphism. We therefore view the biconjugate back on E through this natural identification and write

$$f^{**}(x) := (f^*)^*(\iota_E(x)) = \sup_{\xi \in E^*} \{\langle \xi, x \rangle - f^*(\xi)\} \qquad (x \in E).$$

[1]

Definition 4.2 (Closure and closed extended-value function)**.** Let $f : E \to \mathbb{R} \cup \{+\infty\}$. The *closure* of f, denoted $\mathrm{cl}\, f$, is the function characterized by

$$\mathrm{epi}(\mathrm{cl}\, f) = \mathrm{cl}(\mathrm{epi}(f)).$$

[1] The local Verso page for Lecture 4 contains a one-dimensional conjugate animation: open the Lecture 4 Verso page on the course website.

(a) The function f and its biconjugate f^{**}. (b) Three representative affine lower bounds of f.

Figure 1: One-dimensional illustration of biconjugation for $f(x) = \min\{(x-1)^2+1, (x+1)^2+1\}$, whose biconjugate is $f^{**}(x) = \big(\max\{|x|-1, 0\}\big)^2 + 1$. Thus f^{**} agrees with f on the two outer branches and flattens the middle interval $[-1, 1]$. Panel (b) shows three representative affine lower bounds of f; taking the supremum over all such affine lower bounds recovers f^{**}.

We say that f is *closed* if $\operatorname{epi}(f)$ is a closed subset of $E \times \mathbb{R}$, equivalently if $\operatorname{cl} f = f$. [a]

[a] In finite dimensions, this is equivalent to lower semicontinuity.

Lemma 4.1 (Fenchel–Young inequality and equality case)**.** *Let $f : E \to \mathbb{R} \cup \{+\infty\}$ be proper. Then*

$$\forall x \in E,\ \forall \xi \in E^*, \qquad f(x) + f^*(\xi) \geq \langle \xi, x \rangle .$$

If in addition f is closed and convex, then for every $x \in E$ and every $\xi \in E^*$,

$$f(x) + f^*(\xi) = \langle \xi, x \rangle \iff \xi \in \partial f(x).$$

Proof idea of Lemma 4.1. By definition, $f^*(\xi)$ is the smallest shift that makes the affine function $z \mapsto \langle \xi, z \rangle - f^*(\xi)$ lie below f for every z. Evaluating at $z = x$ gives the inequality, and equality means that this affine lower bound touches f at x, which is exactly the subgradient condition.

Lemma 4.2. *Let $f : E \to \mathbb{R} \cup \{+\infty\}$ be proper. Then $\forall x \in E,\ f^{**}(x) \leq f(x)$.*

This can be read as a weak-duality statement for biconjugation: f^{**} always gives a lower bound on f. Unlike Theorem 4.3, this direction does not require convexity or closedness.

Theorem 4.3 (Fenchel–Moreau)**.** *Let E be a finite-dimensional real normed space, and let $f : E \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex. Then*

$$\forall x \in E, \qquad f^{**}(x) = f(x).$$

More generally, if f is proper and convex, then

$$(\operatorname{cl} f)^* = f^*.$$

Applying Theorem 4.3 to $\operatorname{cl} f$ therefore gives

$$f^{**} = (\operatorname{cl} f)^{**} = \operatorname{cl} f.$$

Proof idea of Theorem 4.3. An affine function lying below f, say $x \mapsto \langle y, x \rangle - a$, is the same thing as a closed halfspace

$$H_{\xi,a} := \{(x,t) \in E \times \mathbb{R} : t \geq \langle \xi, x \rangle - a\}$$

that contains $\operatorname{epi}(f)$. Since f is closed and convex, $\operatorname{epi}(f)$ is a closed convex set, so by Theorem 2.6 it is the intersection of all closed halfspaces that contain it. For fixed x, the smallest height t allowed by that intersection is therefore the supremum of all affine lower bounds evaluated at x, namely

$$\sup_{\xi \in E^*} \{\langle \xi, x \rangle - f^*(\xi)\} = f^{**}(x).$$

This is exactly the height of $\operatorname{epi}(f)$ above x, namely $f(x)$.

4.2 Basic Conjugate Pairs

With the structural theorem in place, it is useful to collect a few conjugate pairs that will recur later in the course.

Example 4.1 (A one-dimensional conjugate pair)**.** Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, and define $f(x) := \frac{|x|^p}{p}$ on $\mathbb{R}$. Then $f^*(\xi) = \frac{|\xi|^q}{q}$ for $\xi \in \mathbb{R}$. Thus

$$\left(\frac{|x|^p}{p}\right)^* = \frac{|\xi|^q}{q}.$$

This is the basic one-dimensional power-law conjugate pair.[a]

[a] Applying Lemma 4.1 to this pair gives $x\xi \leq \frac{|x|^p}{p} + \frac{|\xi|^q}{q}$ for all $x, \xi \in \mathbb{R}$, and in particular $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ for $a, b \geq 0$. This is the classical Young inequality, which explains the 'Young' part of the name 'Fenchel–Young inequality'.

Definition 4.3 (Indicator function)**.** Let $C \subseteq E$. The indicator function of C is the function $\delta_C : E \to \mathbb{R} \cup \{+\infty\}$ defined by

$$\delta_C(x) := \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases}$$

Example 4.2 (Indicator functions and support functions)**.** Let $C \subseteq E$ be nonempty, and define its support function by

$$\sigma_C(\xi) := \sup_{x \in C} \langle \xi, x \rangle \qquad (\xi \in E^*).$$

Then

$$(\delta_C)^* = \sigma_C.$$

In one dimension, if $C = [-1, 1] \subseteq \mathbb{R}$, then

$$\sigma_C(\xi) = \sup_{x \in [-1,1]} \xi x = |\xi|.$$

Thus

$$(\delta_{[-1,1]})^* = |\cdot|.$$

Since $|\cdot|$ is proper, closed, and convex, Theorem 4.3 then gives

$$|\cdot|^* = \delta_{[-1,1]}.$$

So the pair $|x|$ and $\delta_{[-1,1]}$ is the one-dimensional special case of the general indicator/support-function correspondence.

Example 4.3 (Norms and dual unit balls)**.** Let $\|\cdot\|$ be a norm on E, and let $\|\cdot\|_*$ be its dual norm on E^*, defined by

$$\|\xi\|_* := \sup_{\|x\|\le 1} \langle \xi, x\rangle .$$

Then

$$\|\cdot\|^* = \delta_{B_*}, \qquad B_* := \{\xi \in E^* : \|\xi\|_* \le 1\} .$$

Indeed, if $\|\xi\|_* \le 1$, then

$$\langle \xi, x\rangle - \|x\| \le \|\xi\|_*\|x\| - \|x\| \le 0 \qquad \forall x \in E,$$

so $\|\cdot\|^*(\xi) = 0$. If $\|\xi\|_* > 1$, choose $x_0 \in E$ such that $\langle \xi, x_0\rangle > \|x_0\|$; then

$$\langle \xi, tx_0\rangle - \|tx_0\| = t\big(\langle \xi, x_0\rangle - \|x_0\|\big) \to +\infty \qquad (t \to +\infty),$$

so $\|\cdot\|^*(\xi) = +\infty$.

Example 4.4 (Exponential and negative entropy)**.** Let $f(x) = e^x$ on $\mathbb{R}$. Then

$$f^*(\xi) = \begin{cases} \xi \log \xi - \xi, & \xi \ge 0, \\ +\infty, & \xi < 0, \end{cases}$$

with the convention $0 \log 0 := 0$. This is the basic exponential/entropy conjugate pair.

4.3 Marginal Duality and Applications

For subsets $A, B \subseteq Y$, write $A - B := \{a - b : a \in A,\ b \in B\} = A + (-B)$.

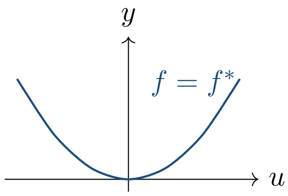
Theorem 4.4 (Marginal duality / perturbation duality)**.** *Let X and U be finite-dimensional real vector spaces, and let*

$$\Phi : X \times U \to \mathbb{R} \cup \{+\infty\}$$

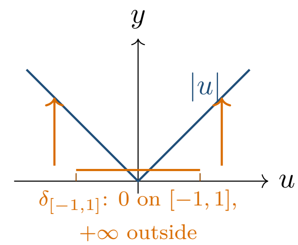
be convex. Define the marginal value function of Φ as

$$p : U \to \mathbb{R} \cup \{\pm\infty\}, \qquad p(u) := \inf_{x \in X} \Phi(x, u).$$

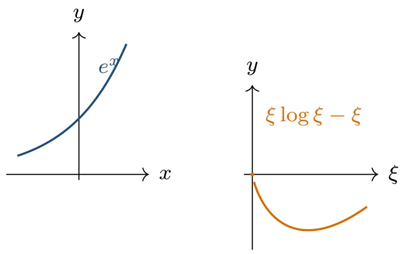
Here p may take the value $-\infty$, even though Φ itself only takes values in $\mathbb{R} \cup \{+\infty\}$. Then the following hold.



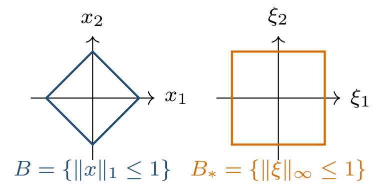
(a) $\frac{1}{2}u^2$ is self-conjugate.



(b) The special case $C = [-1, 1]$, where $\sigma_C(u) = |u|$.



(c) The exponential/negative-entropy pair $e^x \leftrightarrow \xi \log \xi - \xi$.



(d) The norm/dual-ball geometry behind $\| \cdot \|^* = \delta_{B_*}$.

Figure 2: Four basic scalar pictures. Panel (a) shows the self-conjugate quadratic. Panel (b) is the one-dimensional special case $C = [-1, 1]$ of Example 4.2, where the support function is $|u|$. Panel (c) shows the exponential/negative-entropy conjugate pair from Example 4.4. Panel (d) shows the geometric norm/dual-ball picture behind Example 4.3.

1. *For every $y \in U^*$,*

$$p^*(y) = \Phi^*(0, y).$$

Consequently, for every $u \in U$ and every $y \in U^*$,

$$p(u) \ge -\Phi^*(0, y) + \langle y, u \rangle .$$

In particular,

$$\sup_{y \in U^*} \{-\Phi^*(0, y)\} \le p(0).$$

2. *If $p(0) = -\infty$, then*

$$\sup_{y \in U^*} \{-\Phi^*(0, y)\} = -\infty.$$

3. *If $p(0) > -\infty$ and $\partial p(0) \neq \varnothing$, then*

$$p(0) = \max_{y \in U^*} \{-\Phi^*(0, y)\} .$$

In particular, strong duality and dual attainment hold.

4. *Define*

$$D := \{u \in U : \exists x \in X \text{ such that } \Phi(x, u) < +\infty\} = \operatorname{dom} p.$$

If $p(0) \in \mathbb{R}$ and $0 \in \operatorname{ri}(D)$, then

$$p(0) = \max_{y \in U^*} \{-\Phi^*(0, y)\} .$$

One especially useful specialization of Theorem 4.4 is the template

$$\inf_{x \in X} \{f(x) + g(Ax)\}.$$

We postpone that reduction to Exercise 4.1. The worked applications below instead compute the relevant perturbation conjugates directly from Theorem 4.4.

Example 4.5 (Recovering the LP dual)**.** Consider the linear program

$$\inf \left\{ c^\top x : Ax \geq b,\ x \geq 0 \right\},$$

with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$. Define the perturbation

$$\Phi(x, u) := c^\top x + \delta_{\mathbb{R}^n_+}(x) + \delta_{\mathbb{R}^m_+}(Ax - b - u), \qquad (x, u) \in \mathbb{R}^n \times \mathbb{R}^m.$$

Then

$$p(u) := \inf_{x \in \mathbb{R}^n} \Phi(x, u) = \inf \left\{ c^\top x : Ax \geq b + u,\ x \geq 0 \right\}.$$

For $y \in \mathbb{R}^m$,

$$\Phi^*(0, y) = \begin{cases} -b^\top y, & A^\top y \leq c,\ y \geq 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Therefore Theorem 4.4 recovers the dual formula

$$\sup \left\{ b^\top y : A^\top y \leq c,\ y \geq 0 \right\} \leq \inf \left\{ c^\top x : Ax \geq b,\ x \geq 0 \right\},$$

and, if $0 \in \operatorname{ri}(\operatorname{dom} p)$ and the primal value is finite, it gives

$$\inf \left\{ c^\top x : Ax \geq b,\ x \geq 0 \right\} = \max \left\{ b^\top y : A^\top y \leq c,\ y \geq 0 \right\}.$$

This recovers the usual LP dual from the general perturbation template. Lecture 3 proves a sharper strong-duality statement using polyhedral structure, without this extra relative-interior hypothesis.

Proof of Example 4.5. We compute the dual formula in four steps. The main calculation in *Step 2* does not introduce any extra variable; *Step 3* then revisits the same conjugate by introducing a slack variable, because that second route is the one that generalizes most directly to conic programs.

Step 1: compute the perturbation value function $p(u)$. By definition,

$$\Phi(x, u) = c^\top x + \delta_{\mathbb{R}^n_+}(x) + \delta_{\mathbb{R}^m_+}(Ax - b - u).$$

This is finite exactly when

$$x \geq 0 \qquad \text{and} \qquad Ax - b - u \in \mathbb{R}^m_+,$$

that is, exactly when

$$x \geq 0 \qquad \text{and} \qquad Ax \geq b + u.$$

Therefore

$$p(u) := \inf_{x \in \mathbb{R}^n} \Phi(x, u) = \inf \left\{ c^\top x : Ax \geq b + u,\ x \geq 0 \right\}.$$

In particular,

$$p(0) = \inf \left\{ c^\top x : Ax \geq b,\ x \geq 0 \right\}.$$

Step 2 (direct): compute $\Phi^*(0, y)$ without introducing a new variable. Fix $y \in \mathbb{R}^m$. Starting from the definition of the conjugate,

$$\Phi^*(0, y) = \sup_{x \in \mathbb{R}^n,\ u \in \mathbb{R}^m} \left\{ y^\top u - c^\top x - \delta_{\mathbb{R}^n_+}(x) - \delta_{\mathbb{R}^m_+}(Ax - b - u) \right\}.$$

The indicator terms force $x \geq 0$ and $u \leq Ax - b$. Therefore

$$\Phi^*(0, y) = \sup_{x \geq 0} \left\{ \sup_{u \leq Ax - b} y^\top u - c^\top x \right\}.$$

The inner supremum is

$$\sup_{u \leq Ax - b} y^\top u = \begin{cases} y^\top (Ax - b), & y \geq 0, \\ +\infty, & \text{otherwise}, \end{cases}$$

because if $y \geq 0$, the linear form $y^\top u$ is maximized at the largest allowed value $u = Ax - b$, while if some coordinate of y is negative, the corresponding coordinate of u can be sent to $-\infty$. Hence

$$\Phi^*(0, y) = \begin{cases} -b^\top y + \sup_{x \geq 0} \left\{ x^\top (A^\top y - c) \right\}, & y \geq 0, \\ +\infty, & \text{otherwise}. \end{cases}$$

Finally,

$$\sup_{x \geq 0} \left\{ x^\top (A^\top y - c) \right\} = \begin{cases} 0, & A^\top y \leq c, \\ +\infty, & \text{otherwise}, \end{cases}$$

so

$$\Phi^*(0, y) = \begin{cases} -b^\top y, & A^\top y \leq c,\ y \geq 0, \\ +\infty, & \text{otherwise}. \end{cases}$$

Alternative Step 2 (slack variable): the same computation with a slack variable. The direct calculation above is already enough, but it is helpful to see the more structured version that will reappear in conic duality. Starting again from the constraint $Ax - b - u \in \mathbb{R}^m_+$, write

$$s := Ax - b - u \in \mathbb{R}^m_+, \qquad \text{so that} \qquad u = Ax - b - s.$$

Substituting this into the objective gives

$$\begin{aligned} \Phi^*(0, y) &= \sup_{x \geq 0,\ s \geq 0} \left\{ y^\top (Ax - b - s) - c^\top x \right\} \\ &= -b^\top y + \sup_{x \geq 0} \left\{ x^\top (A^\top y - c) \right\} + \sup_{s \geq 0} \left\{ (-y)^\top s \right\}. \end{aligned}$$

Now these two suprema can be read coordinatewise.
For the s-part:

$$\sup_{s \geq 0} \left\{ (-y)^\top s \right\} = \begin{cases} 0, & y \geq 0, \\ +\infty, & \text{otherwise}, \end{cases}$$

because if $y \geq 0$, then $(-y)^\top s \leq 0$ for every $s \geq 0$, with equality at $s = 0$; while if some coordinate of y is negative, one can send the corresponding coordinate of s to $+\infty$.
For the x-part:

$$\sup_{x \geq 0} \left\{ x^\top (A^\top y - c) \right\} = \begin{cases} 0, & A^\top y \leq c, \\ +\infty, & \text{otherwise}, \end{cases}$$

because if $A^\top y \leq c$, then $x^\top (A^\top y - c) \leq 0$ for every $x \geq 0$, with equality at $x = 0$; while if some coordinate of $A^\top y - c$ is positive, one can scale the corresponding basis vector.
Combining the two pieces yields the same formula as above:

$$\Phi^*(0, y) = \begin{cases} -b^\top y, & A^\top y \leq c, \ y \geq 0, \\ +\infty, & \text{otherwise}. \end{cases}$$

Step 3: substitute into Theorem 4.4. Part (1) of Theorem 4.4 gives

$$\sup_{y \in \mathbb{R}^m} \{-\Phi^*(0, y)\} \leq p(0),$$

which becomes

$$\sup \left\{ b^\top y : A^\top y \leq c, \ y \geq 0 \right\} \leq \inf \left\{ c^\top x : Ax \geq b, \ x \geq 0 \right\}.$$

This is exactly the usual weak-duality inequality for the LP pair.
If in addition $0 \in \operatorname{ri}(\operatorname{dom} p)$ and $p(0) \in \mathbb{R}$, then part (4) of Theorem 4.4 upgrades the inequality to equality and gives dual attainment:

$$\inf \left\{ c^\top x : Ax \geq b, \ x \geq 0 \right\} = \max \left\{ b^\top y : A^\top y \leq c, \ y \geq 0 \right\}.$$

□

Example 4.6 (Norm regularization and dual-norm constraints)**.** Let X and Y be finite-dimensional real normed spaces, let $A : X \to Y$ be linear, let $\lambda > 0$, and let $f : X \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex. Consider

$$\inf_{x \in X} \{f(x) + \lambda \|Ax\|\}.$$

Then

$$\inf_{x \in X} \{f(x) + \lambda \|Ax\|\} = \max_{\substack{y \in Y^* \\ \|y\|_* \leq \lambda}} \{-f^*(-A^* y)\}.$$

Proof of Example 4.6. Again, the calculation has three parts.
Step 1: choose the perturbation and identify $p(0)$*.* Set

$$g(u) := \lambda \|u\|, \qquad \Phi(x, u) := f(x) + g(Ax + u).$$

Then the marginal value function is

$$p(u) := \inf_{x \in X} \Phi(x, u) = \inf_{x \in X} \{f(x) + \lambda \|Ax + u\|\}.$$

At $u = 0$, this is exactly the original problem:

$$p(0) = \inf_{x \in X} \{f(x) + \lambda \|Ax\|\}.$$

Because g is finite everywhere on Y, one has $\operatorname{dom} g = Y$. Since f is proper, there exists $x_0 \in X$ with $f(x_0) < +\infty$. For every $u \in Y$,

$$p(u) \leq f(x_0) + \lambda \|Ax_0 + u\| < +\infty,$$

so

$$\operatorname{dom} p = Y, \qquad 0 \in \operatorname{ri}(\operatorname{dom} p).$$

Thus the qualification condition in part (4) of Theorem 4.4 will be automatic as soon as $p(0) \in \mathbb{R}$.
Step 2: compute $\Phi^*(0, y)$. Fix $y \in Y^*$. By definition,

$$\Phi^*(0, y) = \sup_{x \in X,\ u \in Y} \{\langle y, u \rangle - f(x) - g(Ax + u)\}.$$

The standard trick is to replace $Ax + u$ by a new variable:

$$z := Ax + u.$$

Because u ranges freely over all of Y, this change of variables is bijective: for each pair $(x, z) \in X \times Y$, there is a unique $u = z - Ax \in Y$. Thus

$$\begin{aligned}
\Phi^*(0, y) &= \sup_{x \in X,\ z \in Y} \{\langle y, z - Ax \rangle - f(x) - g(z)\} \\
&= \sup_{x \in X,\ z \in Y} \{\langle y, z \rangle - \langle A^* y, x \rangle - f(x) - g(z)\} \\
&= \sup_{x \in X} \{\langle -A^* y, x \rangle - f(x)\} + \sup_{z \in Y} \{\langle y, z \rangle - g(z)\} \\
&= f^*(-A^* y) + g^*(y).
\end{aligned}$$

So the remaining work is to compute g^*.
Step 3: compute g^* *and substitute into the marginal theorem.* For $y \in Y^*$,

$$g^*(y) = \sup_{u \in Y} \{\langle y, u \rangle - \lambda \|u\|\}.$$

If $\|y\|_* \leq \lambda$, then for every $u \in Y$,

$$\langle y, u \rangle \leq \|y\|_* \|u\| \leq \lambda \|u\|,$$

hence

$$\langle y, u \rangle - \lambda \|u\| \leq 0.$$

Taking $u = 0$ shows that the supremum is exactly 0.
If $\|y\|_* > \lambda$, then by the definition of the dual norm there exists $u_0 \in Y$ such that

$$\langle y, u_0 \rangle > \lambda \|u_0\|.$$

Scaling u_0 gives

$$\langle y, tu_0 \rangle - \lambda \|tu_0\| = t(\langle y, u_0 \rangle - \lambda \|u_0\|) \to +\infty \qquad (t \to +\infty),$$

so $g^*(y) = +\infty$. Therefore

$$g^*(y) = \begin{cases} 0, & \|y\|_* \le \lambda, \\ +\infty, & \|y\|_* > \lambda. \end{cases}$$

In other words, the conjugate of the norm penalty is the indicator of the dual norm ball of radius λ.

Now apply Theorem 4.4. If $p(0) = -\infty$, then part (2) already gives

$$\sup_{y \in Y^*} \{-\Phi^*(0, y)\} = -\infty,$$

so the desired identity holds trivially. If instead $p(0) > -\infty$, then $p(0) \in \mathbb{R}$ because $p(0) < +\infty$, and part (4) gives

$$p(0) = \max_{y \in Y^*} \{-\Phi^*(0, y)\}.$$

Substituting the formula for $\Phi^*(0, y)$ and then the formula for $g^*(y)$ produces

$$\inf_{x \in X} \{f(x) + \lambda \|Ax\|\} = \max_{\substack{y \in Y^* \\ \|y\|_* \le \lambda}} \{-f^*(-A^* y)\}.$$

This is the desired dual problem. □

Proofs

Proof of Lemma 4.1. By definition of the conjugate,

$$f^*(y) = \sup_{z \in E} \{\langle y, z \rangle - f(z)\} \ge \langle y, x \rangle - f(x),$$

which rearranges to

$$f(x) + f^*(y) \ge \langle x, y \rangle.$$

Assume now that f is proper, closed, and convex. Equality holds if and only if

$$f^*(y) = \langle y, x \rangle - f(x).$$

By the definition of f^*, this is equivalent to

$$\forall z \in E, \qquad \langle y, z \rangle - f(z) \le \langle y, x \rangle - f(x),$$

that is,

$$\forall z \in E, \qquad f(z) \ge f(x) + \langle y, z - x \rangle.$$

This is exactly the statement that $y \in \partial f(x)$. □

Proof of Example 4.1. By definition,

$$f^*(\xi) = \sup_{x \in \mathbb{R}} \left\{ x\xi - \frac{|x|^p}{p} \right\}.$$

The maximizer has the same sign as ξ, so it is enough to consider $x \ge 0$ and $\xi \ge 0$. For fixed $\xi \ge 0$, set $\varphi_\xi(x) := x\xi - \frac{x^p}{p}$ on $x \ge 0$. Since $\varphi'_\xi(x) = \xi - x^{p-1}$, the unique critical point is

$x = \xi^{1/(p-1)} = \xi^{q-1}$. Substituting gives

$$f^*(\xi) = \xi \cdot \xi^{q-1} - \frac{\xi^{p(q-1)}}{p} = \xi^q - \frac{\xi^q}{p} = \frac{\xi^q}{q}.$$

By symmetry, $f^*(\xi) = \frac{|\xi|^q}{q}$ for all $\xi \in \mathbb{R}$. Applying Lemma 4.1 to this pair then gives the classical Young inequality stated in the footnote. □

Proof of Lemma 4.2. For every $x \in E$ and every $\xi \in E^*$, Lemma 4.1 gives

$$\langle \xi, x \rangle - f^*(\xi) \leq f(x).$$

Taking the supremum over ξ yields

$$f^{**}(x) \leq f(x) \qquad \forall x \in E.$$

□

Proof of Theorem 4.3. By Lemma 4.2, it remains to prove the reverse inequality

$$f(x) \leq f^{**}(x) \qquad \forall x \in E.$$

Fix $x_0 \in E$ and $t_0 < f(x_0)$. Then $(x_0, t_0) \notin \mathrm{epi}(f)$. Because f is proper, closed, and convex, the epigraph $\mathrm{epi}(f)$ is a nonempty closed convex subset of $E \times \mathbb{R}$. By Theorem 2.6, there exists a closed halfspace containing $\mathrm{epi}(f)$ but not (x_0, t_0). Since $\mathrm{epi}(f)$ is upward closed in the t-direction, we may take that halfspace in the form

$$H_{\xi,a} := \{(x, t) \in E \times \mathbb{R} : t \geq \langle \xi, x \rangle - a\}$$

for some $\xi \in E^*$ and $a \in \mathbb{R}$. Thus

$$\mathrm{epi}(f) \subseteq H_{\xi,a} \qquad \text{and} \qquad t_0 < \langle \xi, x_0 \rangle - a.$$

The containment $\mathrm{epi}(f) \subseteq H_{\xi,a}$ is equivalent to saying that the affine function

$$\ell_{\xi,a}(x) := \langle \xi, x \rangle - a$$

is a global affine lower bound for f, i.e.

$$\ell_{\xi,a}(x) \leq f(x) \qquad \forall x \in E.$$

For fixed slope ξ, the smallest admissible intercept is exactly $f^*(\xi)$, because

$$\ell_{\xi,a}(x) \leq f(x) \ \forall x \iff a \geq \sup_{x \in E} \{\langle \xi, x \rangle - f(x)\} = f^*(\xi).$$

Hence every affine lower bound of slope ξ lies below the tight affine function of the same slope,

$$x \mapsto \langle \xi, x \rangle - f^*(\xi).$$

In particular,

$$t_0 < \langle \xi, x_0 \rangle - a \leq \langle \xi, x_0 \rangle - f^*(\xi) \leq f^{**}(x_0).$$

Since this holds for every $t_0 < f(x_0)$, we conclude that

$$f(x_0) \le f^{**}(x_0).$$

Combined with Lemma 4.2, this gives

$$\forall x \in E, \qquad f^{**}(x) = f(x).$$

□

Proof of Theorem 4.4. First we prove that p is convex. Write the strict epigraph of a function h as

$$\operatorname{epi}^{<}(h) := \{(z,t) : h(z) < t\}.$$

For the marginal function p, the strict epigraph is the projection

$$\operatorname{epi}^{<}(p) = \{(u,t) \in U \times \mathbb{R} : \exists x \in X \text{ with } \Phi(x,u) < t\}.$$

Indeed, if $\Phi(x,u) < t$ for some $x \in X$, then $p(u) \le \Phi(x,u) < t$. Conversely, if $p(u) < t$, then the definition of the infimum gives some $x \in X$ with $\Phi(x,u) < t$. The set on the right is the projection of $\operatorname{epi}^{<}(\Phi) \subseteq X \times U \times \mathbb{R}$. Since Φ is convex, $\operatorname{epi}^{<}(\Phi)$ is convex, and projections preserve convexity. Hence $\operatorname{epi}^{<}(p)$ is convex. This implies convexity of p: if $p(u_i) < t_i$ for $i = 1, 2$, then convexity of $\operatorname{epi}^{<}(p)$ gives

$$p(\theta u_1 + (1-\theta)u_2) < \theta t_1 + (1-\theta)t_2, \qquad \theta \in (0,1).$$

Now let $t_i \downarrow p(u_i)$ when $p(u_i) \in \mathbb{R}$. If one value is $+\infty$, the desired convexity inequality is trivial; if one value is $-\infty$, let the corresponding $t_i \to -\infty$. This gives the convexity inequality for p.
Fix $y \in U^*$. By definition,

$$\begin{aligned}
p^*(y) &= \sup_{u \in U} \{\langle y,u\rangle - p(u)\} \\
&= \sup_{u \in U} \sup_{x \in X} \{\langle y,u\rangle - \Phi(x,u)\} \\
&= \sup_{x \in X,\ u \in U} \{\langle 0,x\rangle + \langle y,u\rangle - \Phi(x,u)\} \\
&= \Phi^*(0,y).
\end{aligned}$$

This proves the identity in part (1). For every $u \in U$, the definition of $p^*(y)$ gives

$$p^*(y) = \sup_{v \in U} \{\langle y,v\rangle - p(v)\} \ge \langle y,u\rangle - p(u).$$

Using $p^*(y) = \Phi^*(0,y)$, we obtain

$$p(u) + \Phi^*(0,y) \ge \langle y,u\rangle,$$

which is exactly

$$p(u) \ge -\Phi^*(0,y) + \langle y,u\rangle.$$

Assume now that $p(0) = -\infty$. For every $M > 0$, choose $x_M \in X$ such that

$$\Phi(x_M, 0) \le -M.$$

Then, for every $y \in U^*$,

$$\Phi^*(0, y) \geq \langle 0, x_M \rangle + \langle y, 0 \rangle - \Phi(x_M, 0) \geq M.$$

Since M is arbitrary, $\Phi^*(0, y) = +\infty$ for every y, and hence

$$\sup_{y \in U^*} \{-\Phi^*(0, y)\} = -\infty.$$

This proves part (2).

Assume next that $p(0) > -\infty$ and choose $y \in \partial p(0)$. Then

$$\forall u \in U, \qquad p(u) \geq p(0) + \langle y, u \rangle .$$

Equivalently,

$$\forall u \in U, \qquad \langle y, u \rangle - p(u) \leq -p(0).$$

Taking the supremum over u gives

$$p^*(y) \leq -p(0).$$

On the other hand, evaluating at $u = 0$ gives

$$p^*(y) \geq \langle y, 0 \rangle - p(0) = -p(0).$$

Hence

$$p^*(y) = -p(0).$$

Using part (1) of Theorem 4.4,

$$p(0) = -p^*(y) = -\Phi^*(0, y).$$

Combined with the weak-duality inequality from part (1), this yields

$$p(0) = \max_{v \in U^*} \{-\Phi^*(0, v)\} ,$$

with the maximum attained at the chosen y. This proves part (3).

Finally, define

$$D := \{u \in U : \exists x \in X \text{ such that } \Phi(x, u) < +\infty\} .$$

If $u \in D$, then there exists $x \in X$ such that $\Phi(x, u) < +\infty$, so

$$p(u) = \inf_{x' \in X} \Phi(x', u) \leq \Phi(x, u) < +\infty,$$

hence $p(u) < +\infty$. Conversely, if $p(u) < +\infty$, then not all values $\Phi(x, u)$ can equal $+\infty$; therefore some $x \in X$ satisfies $\Phi(x, u) < +\infty$, and hence $u \in D$. Thus

$$D = \{u \in U : p(u) < +\infty\} .$$

Assume now that $p(0) \in \mathbb{R}$ and $0 \in \mathrm{ri}(D)$. We show that p does not take the value $-\infty$ on D. Fix $u \in D$. Since $p(0) \in \mathbb{R}$, we have $0 \in D$, so $\mathrm{aff}(D)$ is a linear subspace. Because $0 \in \mathrm{ri}(D)$, the set D contains a relative neighborhood of 0 in $\mathrm{aff}(D)$. Therefore there exists $\lambda > 0$ such that $-\lambda u \in D$. If $p(u) = -\infty$, then convexity of p gives

$$p(0) = p\Big(\frac{\lambda}{1 + \lambda} u + \frac{1}{1 + \lambda}(-\lambda u)\Big) \leq \frac{\lambda}{1 + \lambda} p(u) + \frac{1}{1 + \lambda} p(-\lambda u) = -\infty,$$

because $p(-\lambda u) < +\infty$. This contradicts $p(0) \in \mathbb{R}$. Hence $p(u) > -\infty$ for every $u \in D$.

Therefore $p : U \to \mathbb{R} \cup \{+\infty\}$ is a proper convex function, and its usual domain is D. Since $0 \in \mathrm{ri}(D) = \mathrm{ri}(\mathrm{dom}\, p)$, Theorem 2.10 gives $\partial p(0) \neq \varnothing$. Part (3) now applies and proves part (4). $\square$

Exercises

Exercise 4.1. Fenchel–Rockafellar specialization of Theorem 4.4. Let X and Y be finite-dimensional real vector spaces, let $A : X \to Y$ be linear, and let

$$f : X \to \mathbb{R} \cup \{+\infty\}, \qquad g : Y \to \mathbb{R} \cup \{+\infty\}$$

be proper, closed, and convex. Do the following.

1. Define
$$\Phi(x, u) := f(x) + g(Ax + u)$$
and prove that Φ is convex on $X \times Y$.

2. Define
$$p(u) := \inf_{x \in X} \Phi(x, u) = \inf_{x \in X} \{f(x) + g(Ax + u)\}.$$
Show that
$$p(0) = \inf_{x \in X} \{f(x) + g(Ax)\}.$$

3. Prove that for every $y \in Y^*$,
$$\Phi^*(0, y) = f^*(-A^* y) + g^*(y).$$

4. Show that
$$\operatorname{dom} p = \operatorname{dom} g - A(\operatorname{dom} f) = -\big(A(\operatorname{dom} f) - \operatorname{dom} g\big).$$
Deduce that
$$0 \in \operatorname{ri}\big(A(\operatorname{dom} f) - \operatorname{dom} g\big) \iff 0 \in \operatorname{ri}(\operatorname{dom} p).$$

5. Use part (1) of Theorem 4.4 to prove
$$\sup_{y \in Y^*} \{-f^*(-A^* y) - g^*(y)\} \le \inf_{x \in X} \{f(x) + g(Ax)\}.$$

6. Assume additionally that
$$0 \in \operatorname{ri}\big(A(\operatorname{dom} f) - \operatorname{dom} g\big)$$
and that the primal infimum is finite. Use part (4) of Theorem 4.4 to prove
$$\inf_{x \in X} \{f(x) + g(Ax)\} = \max_{y \in Y^*} \{-f^*(-A^* y) - g^*(y)\}.$$

Exercise 4.2. Dual norm twice. Let $\|\cdot\|$ be a norm on E, and let $\|\cdot\|_*$ be its dual norm on E^*. Prove that, under the natural identification $E \simeq E^{**}$, the dual norm of $\|\cdot\|_*$ is the original norm:

$$\|x\|_{**} = \|x\| \qquad \forall x \in E.$$